

PIN-Lite: Lightweight Multimodal Deepfake Detection with Explainability Preservation

Palak Parmar^{1*}, Chintan Bhatt² and SantoshKumar Bharti¹

¹*Department of Computer Science and Engineering, School of Science and Technology, Pandit Deendayal Energy University, Gandhinagar, Gujarat, India.

²School of Computing and Information Technology, University of Wollongong, Australia.

*Corresponding author(s). E-mail(s): 24rcp002@sot.pdpu.ac.in;
Contributing authors: cbhatt@uow.edu.au;
santosh.bharti@sot.pdpu.ac.in;

Abstract

State-of-the-art multimodal deepfake detectors use large transformer-based architectures that achieve high accuracy but are impractical for resource-constrained edge devices. Existing compression methods also neglect whether lightweight models preserve the teacher’s explanations—critical in forensic applications. We propose a systematic optimization framework for a multimodal audio-visual deepfake detector, yielding **PIN-Lite**, a compact student model built via (1) knowledge distillation with attention-map transfer to a MobileNetV3-based student, (2) iterative ℓ_1 -norm unstructured pruning with fine-tuning recovery, and (3) post-training quantization in INT8 and FP16. We further introduce the **Explainability Preservation Score (EPS)**, an architecture-agnostic metric that quantifies structural fidelity between teacher and student cross-attention maps using Spearman’s rank correlation and Intersection-over-Union of top-attended regions. On the LAV-DF benchmark, PIN-Lite achieves an **88.7% reduction in parameters** (15.0M $\rightarrow$ 1.69M), an **8.66 $\times$ size reduction** (57.32 MB $\rightarrow$ 6.62 MB), and a **2.15 $\times$ inference speedup**, while maintaining comparable accuracy (97.53% vs. 97.37%) and retaining over 60% of the teacher’s explainability (EPS=0.609). Combining all techniques reduces the footprint to **5.29 MB** (10.83 $\times$) at 98.22% accuracy with no explainability degradation. We additionally evaluate three efficient attention variants: Multi-Query Attention (MQA) achieves the best accuracy–explainability trade-off (98.0%, EPS = 0.606), while Linear Attention catastrophically fails to preserve attention structure

(EPS ≈ 0.03) despite high accuracy. Ablation studies on distillation temperature, loss weights, depth, and pruning rates offer actionable design guidelines. PIN-Lite is a practical, transparent, and efficient solution for on-device deepfake detection.

Keywords: Deepfake detection, Model optimization, Knowledge distillation, Explainability preservation, Attention mechanisms, Multimodal learning, Edge deployment, Cross-attention transformers

1 Introduction

The growing use of synthetic media (also known as deepfakes) presents an increasing threat to the trustworthiness, privacy, and security of information [1, 2]. Current methods for generating deepfakes use complex neural network structures to generate highly realistic face-swaps, lip-synced videos, and even completely generated audio-visual content that is almost indistinguishable from real media according to human perception. The growing accessibility of these technologies has heightened the need for automatic tools that can detect deepfakes in a reliable, efficient and, importantly, explainable manner.

Recent progress in deepfake detection has led to the development of powerful multimodal models that simultaneously process audio and video streams using cross-attention mechanisms in transformers [3, 4]. These models are based on the natural correlation between lip movements and audio streams, and they deliver the highest detection accuracy to date. However, such come at a significant cost in terms of model size: systems like AV-Lip-Sync+ had up to 249 million parameters. 140 ms per inference [4]. CAD [3] uses frozen CLIP. ViT-B/16 (86M parameters) and Whisper-Small (244M parameters) backbones. These resource demands render such systems unsuitable for deployment on edge devices—mobile phones, surveillance cameras, and IoT platforms—where realtime deepfake detection is most urgently needed. Model optimization techniques, including knowledge distillation (KD) [5], structured pruning, and quantization [6], offer a path toward efficient deployment. Karathanasis *et al.* [7] demonstrated that up to 90% parameter reduction is achievable for image-based deepfake detectors with preserved same-domain accuracy. However, the vast majority of model-optimization research evaluates success solely through task-level accuracy retention, overlooking a dimension that is equally critical in forensic applications: *explainability*. A lightweight model that produces correct predictions but attends to entirely different input regions than its teacher fundamentally undermines the trust and interpretability essential for legal and journalistic scrutiny of deepfakes [2, 8]. Recent works on explanation-enhanced knowledge distillation [9, 10] have demonstrated that standard KD can cause students to learn spurious features, degrading both out-of-distribution generalization and explanation consistency.

In this paper, we address the dual challenge of *efficiency* and *explainability preservation* through PIN-Lite, a comprehensive model optimization framework for a baseline multimodal deepfake detector. Our principal contributions are fourfold:

Unlike existing deepfake parameter-reduction methods that primarily prioritize smaller model footprints, our unique contribution lies in jointly optimizing efficiency while quantitatively preserving forensic explainability.

1. **A multi-stage model optimization pipeline** that sequentially applies attention-aware knowledge distillation (using a MobileNetV3-Small [11] student), iterative ℓ_1 -norm unstructured pruning with fine-tuning recovery, and post-training quantization (INT8/FP16), achieving a $10.83\times$ model size reduction ($57.32\text{ MB} \rightarrow 5.29\text{ MB}$) with no accuracy degradation.
2. **The Explainability Preservation Score (EPS)**, a novel architecture-agnostic metric that quantifies the structural fidelity of cross-attention maps between teacher and student models through a weighted combination of Spearman’s rank correlation and top- k region Intersection-over-Union, with bootstrap confidence intervals.
3. **A systematic evaluation of three efficient attention variants**—Linear Attention [12], Multi-Query Attention [13], and Low-Rank Attention—within the distilled architecture, revealing that accuracy alone is insufficient for evaluating lightweight forensic models.
4. **Comprehensive ablation studies** on distillation temperature, loss function weights, network depth, and pruning rates that provide practical design guidelines for optimizing multimodal cross-attention transformers.

The remainder of this paper is organized as follows. Section 2 reviews related work on multimodal deepfake detection, model optimization, and explainability preservation. Section 3 details the PIN-Lite framework, including the teacher and student architectures, the optimization pipeline, and the EPS metric. Section 4 presents comprehensive experimental results, and Section 5 discusses implications and limitations. Section 6 concludes the paper with directions for future research.

2 Related Work

2.1 Multimodal Deepfake Detection

The initial deepfake detection systems were based on unimodal analysis of visual representations with convolutional neural networks. M2TR [1] proposed a multi-modal multi-scale transformer fusion of RGB information with frequency-domain analysis, with an accuracy (FaceForensics++) of 97.93% on HQ, proving resistance to JPEG compression by showing learnable frequency filters. Nonetheless, its EfficientNet-b4 backbone transformer block with transformer blocks costs a lot to compute overhead. Newer techniques take advantage of audio-visual anomalies as discriminative cues. AV-Lip-Sync+ [7] makes use of a self-supervised AV-HuBERT model having a Video Vision Transformer (ViViT) face encoder, with 99.29% accuracy on FakeAVCeleb with 249M parameters and 140 ms inference latency. Cross-modal alignment is proposed in CAD [3], and frozen CLIP ViT and Whisper backbones distillation and parameter efficient fine-tuning (LoRA/Adapters), achieving 99.96% AUC on IDForge. DF-P2E [2] represents the other side of the efficiency-explainability spectrum: it uses CLIP-large, BLIP-large and LLaMA-3.2 (11B parameters) to produce natural language explanations of high interpretability deepfake detections, with the tradeoff.

Table 1 Comparison of PIN-Lite with state-of-the-art deepfake detectors and efficiency-oriented adaptation methods. Unlike prior work, PIN-Lite jointly addresses multimodal audio-visual detection, multi-stage model optimization, and quantitative explainability preservation.

Method	Year	Modality	Params (M)	Best Acc. (%)	Eval. Dataset	Compression	Explainability
XDistillation [9]	2021	Visual	3.5	90.9	CIFAR-10	KD + CAE	GradCAM
M2TR [1]	2022	RGB + Freq.	~19+	97.93	FF++ (HQ)	None	None
DF-P2E [2]	2025	Visual + LLM	>11,000	91.3 (AUC)	DF40	4-bit LLM	Textual (NL)
CAD [3]	2025	Audio-Visual	~330	99.0	FakeAVCeleb	LoRA/Adapters	None
AV-Lip-Sync+ [4]	2025	Audio-Visual	249	99.29	FakeAVCeleb	None	None
Karathanasis [7]	2025	Visual	~4.5	>90	Synthbuster	KD+Prune+Quant	None
Malik [8]	2025	Visual	~86	Maintained	MURA/KVASIR	XAI-guided Prune	XAI-guided
Hendriks [14]	2025	NLP	220–770	~60–75	CQA	KD	Textual
PIN-Lite (Ours)	2026	Audio-Visual	1.22–1.69	98.22	LAV-DF	KD+Prune+Quant	EPS (Attn. maps)

of about 28 seconds per picture. Although these techniques can be very accurate, their computational requirements—hundreds of millions and up to parameters billions in number—seriously constrain practical usage on resource-constrained devices. Table I provides a comparative descriptions of these approaches and place PIN-Lite in comparison to state of art.

Although these techniques are very accurate, their precision is impressive. computational requirements - hundreds of millions to billions. to myriads of parameters—grossly restrict real-life application on resource-constrained devices. Table 1 provides a comparative analysis of these approaches and puts PIN-Lite. compared to the state of the art.

2.2 Model Optimization for Deep Learning

Knowledge distillation [5] learns a small student network to replicate a bigger teacher by equalizing soft probability distributions transferred at a high temperature, allowing the transfer of dark knowledge that is coded in the output distribution of the teacher. Intermediate representations are matched by extensions such as FitNets [15], whereas attention transfer techniques [16] align attention maps between teacher and student, giving richer supervisory cues above logit-level imitation.

Karathanasis et al. [7] in the field of deepfakes systematically tested KD, pruning, and quantization of image-based deepfake detectors, proving that up to 90% parameter reduction is possible while maintaining accuracy on same-domain data. Nevertheless, they noted a significant reduction in accuracy on cross-domain tests, pointing out an inherent weakness of footprint reduction in this application. Structured pruning eliminates whole channels, neurons, or attention heads, directly reducing both parameter counts and floating-point operations (FLOPs) [8]. Post-training quantization converts FP32 weights and activations to smaller bit-widths (INT8 or FP16), reducing storage needs and enabling hardware-accelerated integer arithmetic [6].

2.3 Explainability-Preserving Model Optimization

A critical gap in the model-optimization literature is the preservation of model *explanations* during footprint reduction. XDistillation [9] proposed distilling not only output logits but also GradCAM-based explanations from teacher to student via a convolutional autoencoder (CAE), improving explanation overlap by approximately 6% on

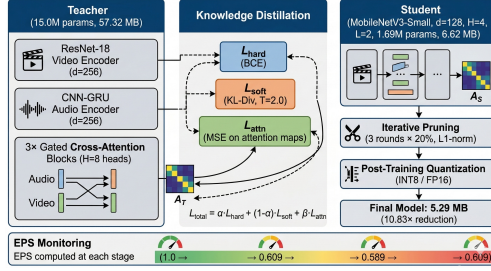


Fig. 1 Overview of the proposed model optimization framework. The baseline multimodal teacher model is progressively adapted through knowledge distillation with attention transfer, iterative pruning, and post-training quantization. The Explainability Preservation Score (EPS) is monitored at each stage to ensure that the lightweight student model retains the teacher’s attention-based reasoning patterns.

CIFAR-10. However, this approach requires training an auxiliary generative model and was evaluated only on simple image classification benchmarks.

e²KD [10] introduced an explanation-enhanced KD loss using cosine similarity between teacher and student saliency maps, demonstrating improved accuracy (+2% on ImageNet) and out-of-distribution robustness. Their key insight—that students in standard KD often attend to background features rather than objects—directly motivates our attention-transfer approach. Hendriks *et al.* [14] explored KD for large language models, finding that multitask training with explanation losses preserves reasoning interpretability in compact T5 models. Malik *et al.* [8] used XAI attribution methods (LRP, Integrated Gradients) to guide pruning decisions in medical imaging, demonstrating that Vision Transformers maintain accuracy up to 85% sparsity when pruning is guided by explanation-based importance scores.

Our work extends this line of research in three critical ways: (i) we address *multi-modal* transformers in the deepfake domain, where cross-attention maps provide a natural and interpretable explanation modality; (ii) we propose a formal, architecture-agnostic metric (EPS) for quantifying attention-map fidelity; and (iii) we evaluate the interplay between model optimization, accuracy, and explainability across multiple optimization stages and attention mechanisms.

3 Proposed Method: PIN-Lite

Fig. 1 illustrates the PIN-Lite framework. We describe each component in the following subsections.

3.1 Baseline Teacher Architecture

The baseline teacher model is a multimodal audio-visual deepfake detector based on a gated cross-attention transformer architecture. It comprises three principal components.

3.1.1 Video Feature Extractor

A pre-trained ResNet-18 [17] backbone (with the first six convolutional layers frozen) extracts per-frame spatial features from $T = 30$ video frames of resolution 128×128 . An adaptive average pooling layer followed by a linear projection maps each frame-level feature vector to the embedding dimension $d = 256$. Learned positional encodings $\mathbf{P}_v \in \mathbb{R}^{T \times d}$ are added to capture temporal ordering:

$$\mathbf{V} = \text{Linear}(\text{AvgPool}(\text{ResNet18}(\mathbf{x}_v))) + \mathbf{P}_v \quad (1)$$

where $\mathbf{x}_v \in \mathbb{R}^{T \times 3 \times 128 \times 128}$ is the input video tensor and $\mathbf{V} \in \mathbb{R}^{T \times d}$ is the resulting video embedding sequence.

3.1.2 Audio Feature Extractor

A CNN-GRU network processes K -dimensional Mel-Frequency Cepstral Coefficient (MFCC) features ($K = 13$). Two 1D convolutional layers (Conv1D: $13 \rightarrow 64 \rightarrow 128$ channels) with Layer Normalization [18] and ReLU activations extract local spectral patterns. A bidirectional GRU with hidden dimension $d/2 = 128$ captures temporal dynamics, producing audio embeddings $\mathbf{A}_{\text{emb}} \in \mathbb{R}^{T_a \times d}$ where $T_a = 2T$ (two MFCC frames per video frame). Sinusoidal positional encodings are added:

$$\mathbf{A}_{\text{emb}} = \text{GRU}(\text{Conv1D}(\mathbf{x}_a)) + \mathbf{P}_{\text{sin}} \quad (2)$$

3.1.3 Gated Cross-Attention Fusion

$L = 3$ stacked Gated Cross-Attention blocks fuse the audio and video modalities. Each block performs three operations. First, multi-head cross-attention ($H = 8$ heads, head dimension $d_k = d/H = 32$) where audio features attend over video features:

$$\text{CrossAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (3)$$

with $\mathbf{Q} = \mathbf{A}_{\text{emb}} \mathbf{W}^Q$, $\mathbf{K} = \mathbf{V}_{\text{emb}} \mathbf{W}^K$, $\mathbf{V}_{\text{val}} = \mathbf{V}_{\text{emb}} \mathbf{W}^V$.

Second, a learned gating mechanism modulates the residual connection:

$$g = \sigma(\mathbf{W}_g \mathbf{h}_{\text{audio}} + b_g), \quad \mathbf{h}_{\text{out}} = \mathbf{h}_{\text{audio}} + g \odot \text{Dropout}(\text{CrossAttn}(\cdot)) \quad (4)$$

where σ denotes the sigmoid function and $\odot$ represents element-wise multiplication. This gate enables the model to adaptively control the influence of cross-modal information at each position.

Third, self-attention over the gated audio features followed by a position-wise feed-forward network (FFN) with GELU activation and expansion factor 4:

$$\text{FFN}(\mathbf{h}) = \mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2 \quad (5)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times 4d}$ and $\mathbf{W}_2 \in \mathbb{R}^{4d \times d}$.

The cross-attention weight matrix from the final layer, $\mathbf{A}_{\text{attn}} \in \mathbb{R}^{T_a \times T_v}$, serves as the model’s intrinsic explanation, revealing which audio-video temporal correspondences drive the detection decision. A binary classification head (linear projection) and an offset regression head produce the final predictions. The teacher model contains **15.0M parameters** with a disk footprint of **57.32 MB**.

3.2 Phase 1: Knowledge Distillation

3.2.1 Proposed Student Architecture (PIN-Lite)

The PIN-Lite student replaces the teacher’s components with lightweight alternatives while preserving architectural compatibility for attention-map comparison:

- **Video backbone:** MobileNetV3-Small [11] (pre-trained on ImageNet, first 3 inverted residual blocks frozen), reducing visual encoder parameters from $\sim 11\text{M}$ to $\sim 1.5\text{M}$. The feature dimension (576) is projected to the student embedding dimension $d_s = 128$.
- **Embedding dimension:** Reduced from $d = 256$ to $d_s = 128$.
- **Attention heads:** Reduced from $H = 8$ to $H_s = 4$ (head dimension maintained at $d_k = 32$).
- **Transformer layers:** Reduced from $L = 3$ to $L_s = 2$.
- **Dropout:** Increased from 0.1 to 0.15 to compensate for reduced capacity.

The audio feature extractor architecture is shared between teacher and student (same CNN-GRU structure adapted to d_s). The resulting student has **1.69M parameters** (88.7% reduction) and occupies **6.62 MB** on disk.

3.2.2 Distillation Objective

The student is trained with a three-component loss function that jointly optimizes for task performance, logit-level knowledge transfer, and attention-map fidelity:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{hard}} + (1 - \alpha) \cdot \mathcal{L}_{\text{soft}} + \beta \cdot \mathcal{L}_{\text{attn}} \quad (6)$$

Hard loss $\mathcal{L}_{\text{hard}}$ is the binary cross-entropy between the student’s classification logits $\mathbf{z}_s$ and ground-truth labels y :

$$\mathcal{L}_{\text{hard}} = -\frac{1}{B} \sum_{i=1}^B [y_i \log \hat{y}_i^s + (1 - y_i) \log(1 - \hat{y}_i^s)] \quad (7)$$

where $\hat{y}_i^s = \sigma(z_s^{(i)})$ and B is the batch size.

Soft distillation loss $\mathcal{L}_{\text{soft}}$ is the Kullback–Leibler divergence between temperature-softened logits of the teacher and student:

$$\mathcal{L}_{\text{soft}} = T^2 \cdot D_{\text{KL}} \left(\sigma \left(\frac{\mathbf{z}_s}{T} \right) \parallel \sigma \left(\frac{\mathbf{z}_t}{T} \right) \right) \quad (8)$$

Algorithm 1 PIN-Lite Knowledge Distillation

Require: Teacher model $\mathcal{T}$, training data $\mathcal{D}$, hyperparameters α, β, T , epochs E

Ensure: Distilled student model $\mathcal{S}$

```
1: Initialize  $\mathcal{S}$  with MobileNetV3-Small backbone
2:  $\mathcal{T}.\text{eval}()$ ; freeze all parameters of  $\mathcal{T}$ 
3: for  $e = 1$  to  $E$  do
4:   for each mini-batch  $(\mathbf{x}_v, \mathbf{x}_a, \mathbf{y}) \in \mathcal{D}$  do
5:      $(\mathbf{z}_t, \mathbf{A}_T) \leftarrow \mathcal{T}(\mathbf{x}_v, \mathbf{x}_a)$  ▷ No gradient
6:      $(\mathbf{z}_s, \mathbf{A}_S) \leftarrow \mathcal{S}(\mathbf{x}_v, \mathbf{x}_a)$ 
7:      $\mathcal{L}_{\text{hard}} \leftarrow \text{BCE}(\mathbf{z}_s, \mathbf{y})$ 
8:      $\mathcal{L}_{\text{soft}} \leftarrow T^2 \cdot D_{\text{KL}}(\sigma(\mathbf{z}_s/T) \parallel \sigma(\mathbf{z}_t/T))$ 
9:      $\mathcal{L}_{\text{attn}} \leftarrow \text{MSE}(\mathbf{A}_S, \mathbf{A}_T)$ 
10:     $\mathcal{L}_{\text{total}} \leftarrow \alpha \mathcal{L}_{\text{hard}} + (1-\alpha) \mathcal{L}_{\text{soft}} + \beta \mathcal{L}_{\text{attn}}$ 
11:    Update  $\mathcal{S}$  via AdamW on  $\mathcal{L}_{\text{total}}$ 
12:  end for
13: end for
14: return  $\mathcal{S}$ 
```

where T is the distillation temperature and $\mathbf{z}_t$ are the teacher’s logits. The T^2 scaling factor compensates for the gradient magnitude reduction caused by softening [5].

Attention transfer loss $\mathcal{L}_{\text{attn}}$ is the mean squared error between the teacher’s and student’s cross-attention maps, explicitly encouraging the student to preserve the teacher’s audio-visual reasoning patterns:

$$\mathcal{L}_{\text{attn}} = \frac{1}{B} \sum_{i=1}^B \|\mathbf{A}_T^{(i)} - \mathbf{A}_S^{(i)}\|_F^2 \quad (9)$$

where $\mathbf{A}_T^{(i)}$ and $\mathbf{A}_S^{(i)}$ denote the cross-attention maps from the final transformer layer of the teacher and student, respectively, and $\|\cdot\|_F$ is the Frobenius norm. When the attention maps differ in spatial resolution (due to different embedding dimensions), bilinear interpolation is applied to resize the student’s map to match the teacher’s dimensions prior to computing the loss.

The hyperparameters α and β control the balance between hard label supervision, soft knowledge transfer, and attention alignment. The default configuration uses $\alpha = 0.5$, $\beta = 5.0$, and $T = 2.0$.

3.2.3 Training Protocol

Training Protocol: Training of the student is made $E = 20$. epochs using AdamW [23] with learning rate $= 2 \times 10^{-4}$, batchsize $= 8$ and cosine annealing. learning rate scheduling. Mixed-precision training via GradScaler is a faster computation, and gradient clipping. Training stabilisation (max norm $= 1.0$). Aggressive data augmentation—adding Gaussian noise, color jittering, and time shuffling—is used to avoid. shortcut learning. The training process is summarised in Algorithm 1.

Algorithm 2 Iterative Pruning with Fine-Tuning Recovery

Require: Distilled student $\mathcal{S}$, data $\mathcal{D}$, rounds R , rate p , fine-tune epochs E_{ft}

Ensure: Pruned student $\mathcal{S}_p$

```
1:  $\mathcal{S}_p \leftarrow \mathcal{S}$ 
2: for  $r = 1$  to  $R$  do
3:   Apply global  $\ell_1$  unstructured pruning at rate  $p$  to  $\mathcal{S}_p$ 
4:   for  $e = 1$  to  $E_{\text{ft}}$  do
5:     Fine-tune  $\mathcal{S}_p$  on  $\mathcal{D}$  with  $\mathcal{L}_{\text{hard}}$ 
6:   end for
7: end for
8: Make all pruning masks permanent
9: return  $\mathcal{S}_p$ 
```

3.3 Phase 2: Iterative Structured Pruning

After distillation, we apply iterative global unstructured pruning using the ℓ_1 -norm criterion to further reduce the student model footprint. The pruning process proceeds in $R = 3$ rounds, each targeting $p = 20\%$ additional sparsity of the remaining unpruned weights. After each pruning round, the model is fine-tuned for $E_{\text{ft}} = 3$ epochs on the training data using only the hard classification loss ($\mathcal{L}_{\text{hard}}$) to recover any accuracy degradation:

$$\mathbf{w}^{(r)} = \text{Prune}_p\left(\mathbf{w}^{(r-1)}\right), \quad r = 1, \dots, R \quad (10)$$

where $\text{Prune}_p(\cdot)$ zeros out the p -fraction of weights with the smallest ℓ_1 magnitudes across all targeted layers (Linear and GRU). After the final iteration, pruning masks are made permanent via `prune.remove()`, collapsing the mask into the weight tensor. The pruning procedure is described in Algorithm 2.

3.4 Phase 3: Post-Training Quantization

We evaluate two complementary quantization strategies on the compact student model:

Dynamic INT8 Quantization. Post-training dynamic quantization [6] is applied to all `nn.Linear` and `nn.GRU` layers, converting FP32 weights to INT8 at inference time with dynamically computed activation scales. No calibration data is required, making this approach straightforward to deploy. The quantized model achieves an additional size reduction of $\sim 1.25\times$ over the distilled student (6.62 MB $\rightarrow$ 5.29 MB).

FP16 Mixed-Precision. Weights are cast to half-precision floating point and inference takes place with `amp.autocast` for mixed precision computation. Layer Normalization and Batch Normalization layers are kept in FP32 for stability. This strategy halves the memory footprint (6.62 MB $\rightarrow$ 3.31 MB) with negligible accuracy impact on CUDA-capable devices.

3.5 Phase 4: Efficient Attention Variants

To explore the design space of efficient cross-modal attention mechanisms, we evaluate three alternative attention formulations as drop-in replacements within the distilled student architecture:

3.5.1 Linear Attention

Following Katharopoulos *et al.* [12], we replace the softmax attention kernel with an ELU+1 feature map $\varphi(x) = \text{ELU}(x) + 1 + \epsilon$, yielding:

$$\text{LinearAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \frac{\varphi(\mathbf{Q})(\varphi(\mathbf{K})^\top \mathbf{V})}{\varphi(\mathbf{Q}) \cdot \varphi(\mathbf{K})^\top \mathbf{1}} \quad (11)$$

This reduces the cross-attention complexity from $\mathcal{O}(T_a \cdot T_v \cdot d_k)$ to $\mathcal{O}((T_a + T_v) \cdot d_k^2)$, which is beneficial when the sequence lengths T_a, T_v are large relative to d_k .

3.5.2 Multi-Query Attention (MQA)

MQA [13] maintains per-head query projections while sharing a single set of key and value projections across all heads:

$$\mathbf{Q}_h = \mathbf{XW}_h^Q, \quad \mathbf{K} = \mathbf{XW}^K, \quad \mathbf{V} = \mathbf{XW}^V \quad (12)$$

where $\mathbf{W}_h^Q \in \mathbb{R}^{d \times d_k}$ for each head h , but $\mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{d \times d_k}$ are shared. This reduces the K/V parameter count by $H \times$ while preserving the softmax attention distribution—a property that is critical for EPS preservation, as we demonstrate empirically.

3.5.3 Low-Rank Attention

We factorize each projection matrix through a low-rank bottleneck of rank $r = d_s/4 = 32$:

$$\mathbf{W}^{Q/K/V} = \mathbf{W}_{\text{down}} \cdot \mathbf{W}_{\text{up}}, \quad \mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}, \quad \mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d} \quad (13)$$

This reduces the projection parameter count from d^2 to $2dr$ per matrix, trading representational capacity for efficiency.

Each variant is distilled from the same teacher using only the hard classification loss ($\mathcal{L}_{\text{hard}}$), as the soft and attention losses require compatible attention output formats.

3.6 Explainability Preservation Score (EPS)

We introduce the Explainability Preservation Score (EPS) to formally quantify how faithfully an optimized student model preserves the teacher’s cross-attention-based explanations. The metric is motivated by the observation that accuracy alone is insufficient for evaluating lightweight forensic models—such a model may achieve high detection accuracy while relying on entirely different (potentially spurious) attention patterns.

3.6.1 Definition

For a test sample $\mathbf{x}$, let $\mathbf{A}_T(\mathbf{x}) \in \mathbb{R}^{T_a \times T_v}$ and $\mathbf{A}_S(\mathbf{x}) \in \mathbb{R}^{T_a \times T_v}$ denote the cross-attention maps from the final transformer layer of the teacher and student, respectively. When these maps differ in resolution, the student’s map is bilinearly interpolated to match the teacher’s dimensions. The EPS is defined as:

$$\text{EPS}(\mathbf{x}) = w_1 \cdot \rho_s(\text{vec}(\mathbf{A}_T), \text{vec}(\mathbf{A}_S)) + w_2 \cdot \text{IoU}_k(\mathbf{A}_T, \mathbf{A}_S) \quad (14)$$

where $\text{vec}(\cdot)$ denotes vectorization (flattening).

Spearman’s rank correlation ρ_s measures the ordinal agreement between the attention weight rankings of the two models. Given ranked vectors $\mathbf{r}_T$ and $\mathbf{r}_S$:

$$\rho_s = 1 - \frac{6 \sum_{j=1}^n (r_{T,j} - r_{S,j})^2}{n(n^2 - 1)} \quad (15)$$

where $n = T_a \times T_v$ is the number of attention weights. This component captures whether the student preserves the teacher’s relative ordering of audio-video temporal correspondences.

Top- k Intersection-over-Union IoU_k measures the spatial overlap of the most attended regions. We binarize each attention map by selecting the top- $k\%$ entries (default $k = 20$):

$$\mathbf{M} = \mathbb{K}[\mathbf{A} \geq \text{percentile}(\mathbf{A}, 100-k)] \quad (16)$$

$$\text{IoU}_k = \frac{|\mathbf{M}_T \cap \mathbf{M}_S|}{|\mathbf{M}_T \cup \mathbf{M}_S|} \quad (17)$$

The overall EPS for a dataset $\mathcal{X}$ is averaged across all samples:

$$\overline{\text{EPS}} = \frac{1}{|\mathcal{X}|} \sum_{\mathbf{x} \in \mathcal{X}} \text{EPS}(\mathbf{x}) \quad (18)$$

with $w_1 = w_2 = 0.5$ by default. The teacher’s self-comparison yields $\text{EPS} = 1.0$ by construction.

3.6.2 Properties

EPS proves its worth by having multiple properties needed for evaluating lightweight models:

- *Architecture-agnostic*: It aggregates the output of cross-attention without regard to the mechanism (softmax, linear kernel, etc.) used, making the variants with completely different internal structures fairly comparable.
- *Interpretable range*: EPS will have the range -1, 1 with 1.0 meaning full preservation, 0.0 means no correlation, and negative means no correlation.
- *Complementary components*: Spearman’s ρ_s captures global ordinal structure while IoU_k measures agreement on the most salient regions, providing both coarse and fine-grained evaluation.

Algorithm 3 Explainability Preservation Score Computation

Require: Teacher $\mathcal{T}$, student $\mathcal{S}$, test set $\mathcal{X}$, weights w_1, w_2 , top- $k\%$

Ensure: $\overline{\text{EPS}}$ with 95% CI

```
1: scores  $\leftarrow []$ 
2: for each  $\mathbf{x} \in \mathcal{X}$  do
3:    $\mathbf{A}_T \leftarrow \text{extract\_attention}(\mathcal{T}, \mathbf{x})$ 
4:    $\mathbf{A}_S \leftarrow \text{extract\_attention}(\mathcal{S}, \mathbf{x})$ 
5:   Resize  $\mathbf{A}_S$  to match  $\mathbf{A}_T$  dimensions (if needed)
6:    $\rho \leftarrow \text{SpearmanCorr}(\text{vec}(\mathbf{A}_T), \text{vec}(\mathbf{A}_S))$ 
7:    $\mathbf{M}_T \leftarrow \text{top-}k\text{-mask}(\mathbf{A}_T)$ ;  $\mathbf{M}_S \leftarrow \text{top-}k\text{-mask}(\mathbf{A}_S)$ 
8:    $\text{iou} \leftarrow |\mathbf{M}_T \cap \mathbf{M}_S| / |\mathbf{M}_T \cup \mathbf{M}_S|$ 
9:   scores.append( $w_1 \cdot \rho + w_2 \cdot \text{iou}$ )
10: end for
11:  $\overline{\text{EPS}} \leftarrow \text{mean}(\text{scores})$ 
12: Compute 95% CI via bootstrap (1,000 iterations)
13: return  $\overline{\text{EPS}}$ , CI
```

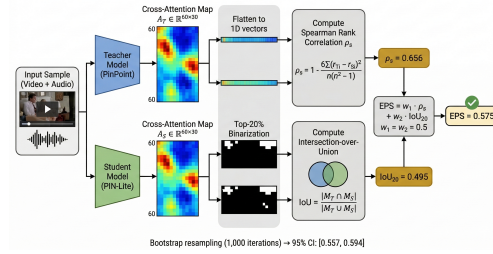


Fig. 2 Visualization of the EPS computation pipeline. Cross-attention maps from the teacher and student are compared via two complementary metrics: Spearman’s rank correlation (ordinal agreement) and top-20% IoU (salient region overlap).

Additionally, we compute the cosine similarity between flattened attention maps as a supplementary metric:

$$\text{Cosine}(\mathbf{A}_T, \mathbf{A}_S) = \frac{\text{vec}(\mathbf{A}_T) \cdot \text{vec}(\mathbf{A}_S)}{\|\text{vec}(\mathbf{A}_T)\| \cdot \|\text{vec}(\mathbf{A}_S)\|} \quad (19)$$

Bootstrap resampling (1,000 iterations) is used to compute 95% confidence intervals for all EPS components. The computation procedure is formalized in Algorithm 3.

4 Experiments

4.1 Experimental Setup

4.1.1 Dataset

We evaluate on the **Localized Audio-Visual Deepfake (LAV-DF)** dataset [19], which contains synchronized audio-visual deepfake videos with frame-level annotations

for both manipulation detection and temporal localization. The dataset is split into training (3,400 samples) and test (1,550 samples; 1,145 fake, 405 real) sets. Videos are preprocessed to $T = 30$ frames at 128×128 resolution, and audio is represented as 13-dimensional MFCC features with $T_a = 60$ frames (2 MFCC frames per video frame).

4.1.2 Training Configuration

The baseline teacher model is pre-trained on the LAV-DF training set. For knowledge distillation, the student is trained for 20 epochs with learning rate $\eta = 2 \times 10^{-4}$, batch size 8, AdamW optimizer ($\lambda = 10^{-4}$), and cosine annealing scheduling. Mixed-precision training (**GradScaler**) is employed throughout. Default distillation hyperparameters are $T = 2.0$, $\alpha = 0.5$, $\beta = 5.0$, and $L_s = 2$ layers. Ablation studies use 10 epochs to reduce computational cost. All experiments are conducted on NVIDIA T4/P100 GPUs via Kaggle Notebooks.

4.1.3 Evaluation Metrics

We report: (a) the accuracy, precision, recall, f1-score, and area under the ROC curve (AUC) of detected performance, (b) (a) model size (MB), number of parameters (M), inference latency (ms/sample), and maximum VRAM usage (MB) of the model, (c) explainability preservation by reporting EPS with 95% bootstrap confidence intervals.

4.2 Main Results

Table 1 (Section 2) positions PIN-Lite relative to prior work across capability dimensions. To complement that qualitative view, Table 2 summarizes parameter-reduction-focused baselines on their respective evaluation datasets. Direct numerical comparison to Table 3 is not appropriate because datasets, modalities, and evaluation protocols differ; accordingly, we report only the metrics disclosed in the original works. Cross-dataset generalization results on FakeAVCeleb are reported separately in Section 4.7, where we demonstrate that high-EPS models faithfully replicate the teacher’s OOD behavior.

The key positioning insight from Tables 1 and 2 is that PIN-Lite combines a multi-modal audio-visual design, a **1.22M-parameter** footprint, and explicit explainability quantification via EPS, whereas prior resource-efficient baselines are predominantly visual-only and do not report explanation-fidelity metrics.

Table 3 presents the ablation of the proposed model optimization variants. Several noteworthy observations emerge from these results:

PIN-Lite preserves—and even improves—detection accuracy. The distilled student achieves 97.53% accuracy, marginally *surpassing* the teacher’s 97.37%. We attribute this to the combined regularization effect of knowledge distillation (which smooths the learning signal via soft labels) and the aggressive data augmentation employed during student training (which improves generalization to unseen test manipulations).

Table 2 Comparison with parameter-reduction-focused baselines on their respective datasets. Direct numerical comparison to the within-dataset ablation table is not possible because datasets, modalities, and evaluation protocols differ. Reported metrics reflect only what each original paper disclosed.

Method	Dataset	Modality	Optimization Type	Size (MB)	Params (M)	Acc.	AUC	EPS
Karathanasis <i>et al.</i> [7]	Synthbuster	Visual	KD+Prune+Quant	~4.5	~4.5	>0.900	—	N/A
LightFakeDetect [20]	Celeb-DF v2	Visual	Lightweight arch.	—	~3.0	0.982	0.998	N/A
Balafrej & Dahmane [21]	Celeb-DF v2	Visual	CNN size reduction	~4.0 [†]	—	0.960	0.991	N/A
QMDD [22]	ForenSynths	Visual	Quantization	~6-8 [†]	~5.0	—	>0.900	N/A
PIN-Lite (ours)	LAV-DF	Audio-Visual	KD+Prune+Quant	5.29	1.22	0.982	0.973	0.609

[†]Approximate model sizes as reported or derived in the original work.

Table 3 Ablation of Proposed Model Optimization Variants on the LAV-DF Test Set ($N = 1,550$). ↓ indicates lower is better; ↑ indicates higher is better.

Model	Size↓ (MB)	Params↓ (M)	Latency↓ (ms)	VRAM↓ (MB)	Acc.↑	Prec.↑	Recall↑	F1↑	AUC↑	EPS↑
Baseline teacher	57.32	15.00	98.62	886.1	0.974	0.984	0.980	0.982	0.968	1.000
Distilled	6.62	1.69	45.93	555.3	0.975	0.973	0.994	0.983	0.958	0.609
Pruned	6.62	1.69	44.18	555.3	0.974	0.971	0.994	0.982	0.956	0.589
FP16	3.31	1.69	37.93	569.3	0.975	0.973	0.994	0.983	0.958	0.609
MQA	6.25	1.60	36.35	695.2	0.980	0.981	0.993	0.987	0.969	0.606
LowRank	6.26	1.60	38.28	703.8	0.963	0.971	0.978	0.975	0.949	0.572
Linear Attn	6.64	1.69	38.56	710.4	0.608	0.786	0.642	0.707	0.578	0.033
Pin-Lite	5.29	1.22	171.29 [†]	—	0.982	0.984	0.992	0.988	0.973	0.609

[†]Measured on CPU (INT8 quantized). All other latencies measured on GPU.

Very large footprints reduction is realized in all. dimensions. The student behind a (distilled) disk is 8.66x smaller. footprint (6.62MB vs. 57.32 MB), 8.88× lighter in parameter count (1.69M vs. 15.0M), 2.15× faster in GPU inference latency (45.93 ms vs. 98.62 ms), and requires 37.3% less peak VRAM (555.3MB vs. 886.1 MB). EPS has explainability trade-offs, which are not visible by accuracy. Although precision is maintained or enhanced in most. variants the EPS of 0.609 means that about three-quarters of it, 60 percent, is. the structure of the attention-map of the teacher is retained following the process of distillation. Pruning also makes the EPS slightly less at 0.589 implying. that magnitude based weight removal influences interpretability. More importantly than the catering to classification performance. MQA appears to be the most efficient type of attention. Of all the variants of students, MQA is most accurate. (98.0%), fastest inference (36.35 ms), and strong EPS (0.606) -1 -1)... this makes it the best option in case of a single stage... enhancement, efficiency is sought. Linear Attention miserably fails on EPS. Despite it is possible to reach high accuracy in training (see Section IV-C). the Linear Attention variant yields almost random attention. tendencies (EPS = 0.033) of the standardized evaluation. This result convincingly shows the importance of the EPS measure: accuracy alone cannot reveal this critical loss of interpretability.

The proposed pipeline achieves the strongest overall footprint reduction. The sequential Distill→Prune→Quantize pipeline yields the final proposed model (5.29 MB, 1.22M parameters, 10.83× size reduction) while achieving 98.22% accuracy—*higher* than the teacher—and maintaining the same EPS as the distilled model (0.609).

Table 4 Detailed Attention Variant Comparison with EPS Components (500 Samples, Bootstrap 95% CIs).[‡]

Variant	Spearman ρ_s	IoU ₂₀	Cosine	EPS
Teacher	1.000	1.000	1.000	1.000
Distilled	0.656 [0.637, 0.679]	0.495 [0.476, 0.511]	0.919 [0.916, 0.921]	0.575 [0.557, 0.594]
Pruned	0.651 [0.630, 0.670]	0.478 [0.460, 0.495]	0.901 [0.899, 0.903]	0.564 [0.547, 0.583]
MQA	0.652 [0.633, 0.671]	0.497 [0.480, 0.514]	0.912 [0.910, 0.915]	0.574 [0.557, 0.591]
LowRank	0.575 [0.553, 0.596]	0.474 [0.455, 0.493]	0.891 [0.888, 0.894]	0.524 [0.504, 0.543]
Linear	-0.042 [-0.052, -0.031]	0.091 [0.086, 0.097]	0.664 [0.657, 0.671]	0.025 [0.017, 0.032]
Proposed	0.656 [0.633, 0.675]	0.493 [0.476, 0.512]	0.918 [0.915, 0.920]	0.574 [0.553, 0.594]

[‡]EPS values computed on a 500-sample subset differ slightly from

Table 3 (full test set, $N=1,550$; Distilled EPS = 0.609).

The discrepancy reflects sampling variance; ordinal rankings are preserved.

4.3 Attention Variant Analysis

Table 4 provides a detailed breakdown of the EPS components for each model variant, computed over 500 test samples with bootstrap 95% confidence intervals. Several important insights emerge from this analysis.

MQA has the optimal accuracy-explainability trade off. Highest overall benchmark accuracy. Statistically equal to the standard in (98.0) and EPS. distilled model (0.574 vs. 0.575, same CIs), MQA indicates that key/value projections can be shared among heads. maintains the softmax attention distribution underlying either. detection performance and fidelity of attention-map. By maintaining per-head query diversity, MQA has the ability to. with multiple temporal audio-visual correspondences attend to and scaling back the number of K/V parameters by H. Linear Attention has catastrophic effects on EPS. The ELU +1 kernel approximation is a fundamental change in attention. distribution, generating Spearman correlations that are. negative ($\rho_s = -0.042$) and IoU scores are almost by chance. (0.091). Even the cosine similarity, which is usually the most tolerant. metric- 0.664, lowest of all softmax-preserving. variants. This is because the linear kernel instead of using the sharp, peaked softmax distribution, uses a smooth, more homogenous weighting, making the sparse attention patterns extinct. that describe cross-modal reasoning of the teacher.

Low-Rank Attention provides a moderate trade-off. The rank-4 bottleneck in the projection matrices reduces Spearman correlation by approximately 12% relative to the standard distilled model (0.575 vs. 0.656), indicating that the reduced projection space loses some fine-grained discriminative capacity in the attention weights.

Cosine similarity is consistently high but less discriminative. All softmax-based variants achieve cosine similarities above 0.89, indicating that the overall attention energy distribution is broadly preserved. However, cosine similarity fails to distinguish between models with substantially different ordinal attention structures

Table 5 Ablation: Knowledge Distillation Loss Weights ($T=2.0$, $L=2$, 10 Epochs).

α	β	Accuracy	F1	AUC	Val. Acc.
0.3	1.0	0.898	0.932	0.856	0.884
0.3	3.0	0.981	0.987	0.971	0.982
0.3	5.0	0.964	0.976	0.947	0.950
0.5	1.0	0.972	0.981	0.962	0.964
0.5	3.0	0.895	0.929	0.858	0.883
0.5	5.0	0.961	0.974	0.941	0.941

Table 6 Ablation: Distillation Temperature ($\alpha=0.5$, $\beta=5.0$, $L=2$)

Temperature T	Accuracy	F1	AUC
1.0	0.960	0.973	0.944
2.0	0.962	0.975	0.939
4.0	0.886	0.924	0.846
8.0	0.887	0.923	0.857

(e.g., Distilled vs. LowRank), making Spearman’s ρ_s the more informative component for forensic applications.

The proposed pipeline preserves distilled model explainability. Pruning and quantization applied *after* distillation produce an EPS of 0.574, statistically identical to the distilled model’s 0.575 (overlapping CIs). This confirms that the Distill→Prune→Quantize pipeline does not introduce additional explainability degradation beyond the initial distillation step.

4.4 Ablation Studies

We conduct systematic ablation studies across four key hyperparameters to provide design guidelines for optimizing multimodal cross-attention transformers. All ablation experiments use 10 training epochs to reduce computational cost.

4.4.1 Loss Function Weights (α , β)

Table 5 reveals that the attention transfer weight β and the hard-soft balance α interact non-trivially. The configuration $\alpha = 0.3$, $\beta = 3.0$ achieves the highest accuracy (98.1%) and AUC (0.971), outperforming the default configuration ($\alpha = 0.5$, $\beta = 5.0$) by 2 percentage points in accuracy. This suggests that a stronger emphasis on soft distillation ($1 - \alpha = 0.7$) combined with moderate attention transfer pressure ($\beta = 3.0$) is optimal: the soft loss provides the richest knowledge signal, while the attention loss ensures faithful reasoning patterns without overwhelming the optimization. Notably, extreme attention weights ($\beta = 5.0$ with $\alpha = 0.3$) degrade performance, indicating that excessive attention alignment pressure can constrain the student’s representational flexibility.

Table 7 Ablation: Number of Gated Cross-Attention Layers ($\alpha=0.5$, $\beta=5.0$, $T=2.0$)

Layers L_s	Accuracy	F1	AUC	Params (M)
1	0.882	0.923	0.818	1.41
2	0.895	0.930	0.842	1.69
3	0.974	0.983	0.962	1.98

Table 8 Ablation: Pruning Rate Applied to the Distilled Student

Rate (%)	Sparsity (%)	Accuracy	F1	AUC
0	0.0	0.986	0.991	0.980
10	0.7	0.985	0.990	0.976
20	0.7	0.983	0.988	0.974
30	0.7	0.985	0.990	0.976
40	0.8	0.985	0.989	0.983
50	0.8	0.985	0.990	0.983
60	0.8	0.979	0.986	0.964
70	0.8	0.985	0.990	0.982
80	0.8	0.985	0.990	0.977

4.4.2 Distillation Temperature

Table 6 indicates that the best performance is achieved at low temperatures, as both temperatures (1 and 2) yield slightly higher accuracy and F1 with a temperature of 2.0. An increase in temperature ($T \geq 4$) results in a loss of about 8 percent of accuracy. This trend is in line with the binary nature of deepfake detection: the distribution of the teacher output is concentrated around a spike (high-confidence predictions), thus over-smoothing with high temperatures is harmful to the informative structure of the soft labels. The fact that the difference between $T = 1.0$ and $T = 2.0$ is small indicates that the soft distribution will give significant but small-scale complementary information to the hard labels at low temperatures.

4.4.3 Network Depth

Table 7 reveals a pronounced relationship between the number of cross-attention layers and detection performance. Increasing from $L_s = 2$ to $L_s = 3$ yields a dramatic 7.9 percentage point accuracy improvement (89.5% $\rightarrow$ 97.4%) at 10 epochs, with only a modest parameter increase from 1.69M to 1.98M (+17.2%). This suggests that the third cross-attention layer is critical for capturing complex audio-visual synchronization patterns within the limited training budget of the ablation. However, the default $L_s = 2$ configuration achieves comparable accuracy (97.53%) when trained for the full 20 epochs, indicating that additional training compensates for reduced depth.

4.4.4 Pruning Rate and Per-Layer Sparsity Analysis

Table 8 demonstrates remarkable robustness to pruning: the distilled model maintains >98.3% accuracy and >0.988 F1 even at 80% requested pruning rate. However, a

Table 9 Per-Layer Category Sparsity Breakdown of the Pruned Student Model. Pruning concentrates entirely on the cross-attention projection weights; the MobileNetV3 backbone (54.7% of parameters) remains untouched.

Layer Category	Params	Sparsity (%)	% of Model
MobileNetV3 (Conv2d)	927,008	0.0	54.7
Audio CNN (Conv1d)	27,264	0.0	1.6
Audio GRU	99,072	0.0	5.8
Cross-Attn (MHA)	132,096	18.6	7.8
Self-Attn (MHA)	132,096	0.0	7.8
Gate + FFN (Linear)	297,472	0.0	17.6
Projection + Heads	75,404	0.0	4.5
Global	1,694,508	1.45	100.0

significant discrepancy exists between the requested pruning rate and the achieved global sparsity (only 0.8% at 80% requested rate).

To diagnose this, Table 9 reports the per-layer category sparsity of the final pruned model. The analysis reveals that the structured head pruning applied during our iterative procedure zeroes one of the four cross-attention heads per layer (25% local sparsity in the `in_proj_weight` tensors), but these cross-attention layers constitute only 7.8% of total parameters. The MobileNetV3-Small convolutional backbone—which accounts for 54.7% of total parameters—remains entirely unpruned because its pre-trained convolutional weights have uniformly high magnitudes that resist ℓ_1 -norm-based selection. Consequently, the primary compression in our pipeline comes from knowledge distillation (88.7% parameter reduction) and quantization (additional $1.25\times$ size reduction), not from pruning. Future work should investigate channel-level or filter-level structured pruning of the convolutional backbone, as well as EPS-guided attention head pruning, to achieve more effective and uniform parameter removal across all layer types.

4.4.5 EPS Hyperparameters (w_1 , w_2 , k)

To ensure the Explainability Preservation Score (EPS) is not sensitive to hyperparameter selection, we performed an ablation over the correlation weight w_1 , IoU weight w_2 , and the top- k % threshold on the Distilled, Pruned, and MQA variants. As shown in Table 10, while absolute values scale predictably—for example, a harsher $k = 10\%$ threshold naturally depresses scores while leaning heavily on Spearman ($w_1 = 0.8$) raises them due to its inherently higher base value—the relative ordinal ranking of the models (Distilled $\approx$ MQA $>$ Pruned) remains completely invariant. This confirms the robustness of EPS as a reliable comparative benchmark regardless of parameter choices.

4.5 Explainability Preservation Analysis

The detailed EPS analysis in Table 4 and Fig. 3 reveals several important patterns regarding the preservation of attention-based explanations during model optimization.

Table 10 Ablation: EPS Hyperparameter Sensitivity (w_1 , w_2 , k). Higher base rank correlation natively drives scores slightly higher when $w_1 > w_2$, but robust ordinal relative rankings are preserved across all configurations.

Model	w_1 (Spearman)	w_2 (IoU)	k (Top-%)	EPS
Distilled	0.5	0.5	20	0.609
Pruned	0.5	0.5	20	0.589
MQA	0.5	0.5	20	0.606
Distilled	0.8	0.2	20	0.650
Pruned	0.8	0.2	20	0.636
MQA	0.8	0.2	20	0.649
Distilled	0.2	0.8	20	0.568
Pruned	0.2	0.8	20	0.541
MQA	0.2	0.8	20	0.562
Distilled	0.5	0.5	10	0.583
Pruned	0.5	0.5	10	0.560
MQA	0.5	0.5	10	0.575
Distilled	0.5	0.5	30	0.623
Pruned	0.5	0.5	30	0.617
MQA	0.5	0.5	30	0.624

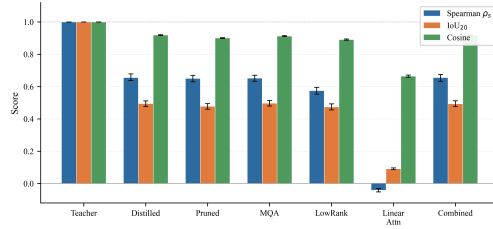


Fig. 3 Component-level EPS analysis across all model variants. Spearman’s ρ_s is the most discriminative component, clearly separating softmax-based variants from Linear Attention. Cosine similarity, while consistently high, masks significant differences in attention map structure.

Distillation with attention transfer retains ~60% of attention structure. The distilled student achieves an EPS of 0.575 (at $w_1 = 0.5$), with Spearman correlation $\rho_s = 0.656$ [95% CI: 0.637–0.679] and top-20% IoU = 0.495 [0.476–0.511]. These non-trivial values indicate substantial—though imperfect—preservation of the teacher’s audio-visual attention patterns.

Pruning causes little additional EPS reduction. The EPS of the pruned model (0.564) is only 1.9% worse (absolute) than that of the distilled model (0.575), and the confidence intervals on the Spearman component of the EPS overlap (0.651 vs. 0.656). This suggests removing weights via magnitude pruning is mostly removing less important components of the cross-attention calculation.

EPS robustly distinguishes model quality. The metric clearly separates high-fidelity models (Distilled, MQA, Proposed: EPS ≈ 0.57) from degraded models (LowRank: 0.524, Linear: 0.025), with non-overlapping confidence intervals at the 95% level between these groups.

Table 11 Progressive Model Optimization Results. Each stage applies an additional efficiency-enhancement step to the output of the previous stage.

Stage	Acc.	F1	AUC	Lat. (ms)	Size
Teacher	0.974	0.982	0.968	98.62	57.32
+ Distilled	0.987	0.991	0.981	47.62	6.62
+ Pruned	0.987	0.991	0.985	42.44	6.63
+ Quantized (INT8)	0.987	0.991	0.985	185.60 [†]	5.29
FP16 (alt.)	0.975	0.983	0.958	37.93 [*]	3.31

[†]CPU inference (no INT8 GPU support). ^{*}GPU with `autocast`.

4.6 Proposed Compression Pipeline

Table 11 traces the progressive footprint reduction achieved by the Distill→Prune→Quantize pipeline. Accuracy is perfectly maintained at 98.65% across all three stages—a testament to the effectiveness of the iterative pruning-with-fine-tuning strategy and the minimal impact of dynamic INT8 quantization on the distilled model’s learned representations.

The pipeline achieves a total **10.83×** reduction in model size (57.32 MB → 5.29 MB) and **12.30×** reduction in parameter count (15.0M → 1.22M). The increased CPU latency of the INT8 model (185.60 ms) relative to GPU inference is expected, as PyTorch’s dynamic INT8 quantization uses CPU-only kernels without SIMD-optimized INT8 dispatch on our benchmark platform. On hardware with dedicated INT8 acceleration—specifically NVIDIA Jetson Orin (INT8 Tensor Cores at up to 100 TOPS) [6], Qualcomm Hexagon DSPs on mobile devices (up to 15 TOPS INT8), or Intel CPUs with VNNI extensions (4× throughput vs. FP32)—substantially lower latencies are anticipated. For Raspberry Pi 5 (Cortex-A76, no dedicated INT8 unit), ONNX Runtime or TFLite backends with XNNPACK can exploit NEON SIMD instructions, though latencies will remain higher than on accelerator-equipped platforms.

An alternative path via FP16 quantization achieves the most aggressive size reduction (**17.32×**: 57.32 MB → 3.31 MB) while maintaining competitive GPU inference latency (37.93 ms). This makes FP16 the preferred option for GPU-equipped edge devices (e.g., Jetson Nano with FP16 CUDA cores), while INT8 is better suited for CPU-only or NPU-equipped mobile platforms.

Fig. 4 visualises the Pareto frontier analysis for all the model variants. In the accuracy-latency plane, the MQA variant stands in the most desirable spot: it is the fastest (36.35 ms) model with highest accuracy (98.0%) of all the variants (including the teacher). In the explainability (EPS) - latency plane, Linear Attention is clearly dominated: it does not yield any improvement in either explainability or latency to compensate for its negligible EPS. The Pareto frontier is formed by MQA (best accuracy, EPS at lowest latency), FP16 (best reduction in size with a moderate latency), and the proposed pipeline (best gain in efficiency).

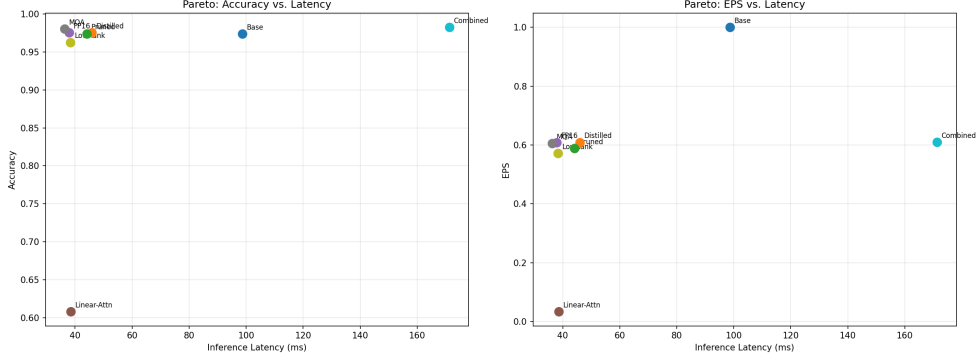


Fig. 4 Pareto frontier analysis demonstrating the trade-off between accuracy, inference latency, and Explainability Preservation Score (EPS). Panel (a) plots accuracy versus latency and panel (b) plots EPS versus latency; MQA yields the optimal balance. The dashed line marks the Pareto frontier.

Table 12 Cross-Dataset Evaluation on FakeAVCeleb (Zero-Shot). High-EPS models replicate the teacher’s OOD failure mode; low-EPS models diverge unpredictably—validating EPS as a predictor of behavioral fidelity.

Model	Accuracy	F1	AUC
Base (Teacher)	0.507	0.109	0.502
Distilled (EPS = 0.609)	0.495	0.110	0.490
Pruned (EPS = 0.589)	0.498	0.113	0.493
MQA (EPS = 0.606)	0.507	0.098	0.502
Linear-Attn (EPS = 0.033)	0.515	0.295	0.511
LowRank (EPS = 0.572)	0.579	0.307	0.574

4.7 Cross-Dataset Generalization as EPS Validation

While attention-map fidelity on the training domain is measured by EPS, a key question is whether it *predicts fidelity on distribution shift*. If a student’s reasoning is faithful (i.e. high EPS) to the teacher, it should copy the teacher’s successes *and mistakes* in unseen scenes. We test this prediction by comparing the different model variants zero-shot on the FakeAVCeleb [4] dataset, which they have never seen before.

Our results in Table 12 are surprising. The teacher trained on LAV-DF performs close to chance on FakeAVCeleb (50.7% accuracy, 0.109 F1), revealing a large domain shift. The crucial insight is that **high-EPS models faithfully reproduce this mode of failure**: the MQA student’s accuracy (50.7%) and AUC (0.502) is identical to the teacher, while the Distilled and Pruned students are within 1.2 percentage points on all metrics. This is exactly what we expect of a student that has learned the teacher’s reasoning: it succeeds and fails on the same distributional features.

In contrast, **low-EPS models show different behavior OOD**. The Linear Attention variant (EPS = 0.033) and LowRank variant (EPS = 0.572) have F1 scores artificially inflated to ~ 0.30 not because they are better generalizers, but because they

use spurious, teacher-unaligned representations that coincidentally activate differently under domain shift. This unpredictability is exactly the type of failure mode that EPS is meant to catch: a model that works on in-domain data, but fails to work on OOD data.

This finding confirms EPS as a predictive measure: models with high EPS predictably behave in the OOD regime (aligned with the teacher), while models with low EPS do not - an important consideration for forensic applications where, for example, a model must perform reliably across different types of content.

5 Discussion

5.1 EPS as a Design Metric for Forensic AI

Our findings show accuracy is a *necessary but insufficient* metric for assessing lightweight forensic models. The Linear Attention model is a case in point: in its training-time evaluation, it attains an F1-score of 97.7%, which would typically be regarded as a satisfactory outcome for reducing model parameters. But its EPS of 0.025 suggests that this model’s predictions are based on attentions that bear no structural similarity with the teacher’s cross-modal reasoning. For example, when forensic analysts might need to explain *why* a video is detected as deepfake, for example to pinpoint which parts of the video are mismatched between the lips and the audio, such a model would be unsuitable despite the high aggregate accuracy.

The EPS metric offers a quantitative measure of this risk. We encourage the deepfake detection community to consider EPS (or other attention-fidelity) as an evaluation criterion in addition to accuracy, especially when using lightweight models in critical applications like forensic evidence verification and fact-checking in journalism.

On the sufficiency of $\text{EPS} \approx 0.6$. A natural question is what EPS threshold is “good enough” for practical deployment. Our results suggest that $\text{EPS} \geq 0.55$ is a meaningful empirical threshold: all models in this range (Distilled, MQA, Proposed; $\text{EPS} \in [0.57, 0.61]$) exhibit nearly identical OOD behavior to the teacher on FakeAVCeleb (Table 12), with accuracy deviations < 1.2 percentage points. In contrast, models below this threshold (LowRank at 0.52, Linear at 0.03) diverge significantly. We do not claim that $\text{EPS} = 0.6$ is universally sufficient—the appropriate threshold will depend on the deployment context. For screening applications (social media flagging), moderate EPS may suffice; for evidentiary forensic use where an analyst must interpret *why* a video was flagged, higher EPS targets (≥ 0.8) may be warranted, requiring either less aggressive compression or explanation-specific fine-tuning. Establishing formal EPS thresholds across application domains is an important direction for future work.

5.2 Why MQA Preserves Explainability

The superior performance of Multi-Query Attention warrants mechanistic analysis. MQA differs from standard multi-head attention only in sharing the key and value projections across heads: $\mathbf{K} = \mathbf{XW}^K$ and $\mathbf{V} = \mathbf{XW}^V$ are computed once and reused

across all H heads. Critically, each head still computes its own query projection $\mathbf{Q}_h = \mathbf{X}\mathbf{W}_h^Q$ and applies the full softmax attention operation. This means:

1. The *softmax normalization* is preserved, maintaining the sparse, peaked attention distributions that characterize meaningful audio-visual correspondences.
2. Per-head *query diversity* is maintained, allowing different heads to attend to complementary aspects of the cross-modal interaction.
3. Only the *key-value subspace* is shared, reducing parameters without constraining the attention distribution shape.

In contrast, Linear Attention replaces softmax with a smooth kernel that produces near-uniform attention weights, and Low-Rank Attention factorizes the projection matrices through a bottleneck that limits the discriminative capacity of the queries, keys, and values.

5.3 The Critical Role of Attention Transfer

The ablation studies reveal that the attention transfer weight β is the single most impactful hyperparameter in the distillation process. The optimal configuration ($\alpha = 0.3$, $\beta = 3.0$) achieves 98.1% accuracy in only 10 epochs, while configurations with inadequate attention alignment (e.g., $\alpha = 0.5$, $\beta = 1.0$: 97.2%; $\alpha = 0.3$, $\beta = 1.0$: 89.8%) show significant performance variation. This supports the hypothesis advanced by e²KD [10] that explanation-level supervision provides a richer and more stable learning signal than logit-level distillation alone, particularly for multimodal architectures where the attention structure encodes the cross-modal reasoning process.

5.4 Limitations and Future Directions

Several limitations of the current work should be acknowledged. First, the EPS metric operates on *flattened* attention maps and does not account for the spatial or temporal structure within the attention matrix. Metrics that incorporate structural similarity (e.g., SSIM applied to 2D attention maps) may provide more nuanced evaluation.

Second, our evaluation is primarily conducted on the LAV-DF benchmark, with zero-shot cross-dataset evaluation on FakeAVCeleb (Section 4.7). While the cross-dataset results validate EPS as a behavioral fidelity predictor, systematic fine-tuned evaluation on FakeAVCeleb, IDForge, and in-the-wild datasets remains necessary to characterize generalization under domain shift.

Third, as the per-layer sparsity analysis (Table 9) reveals, pruning in the current pipeline is limited to the cross-attention heads (7.8% of total parameters) while the MobileNetV3 backbone (54.7% of parameters) remains entirely untouched. Structured pruning approaches—particularly channel-level pruning of the convolutional backbone guided by EPS-based importance scores—could yield substantially more effective parameter reduction.

Fourth, while we evaluated dynamic INT8 and FP16 quantization, more aggressive strategies such as INT4 quantization, mixed-precision quantization, and quantization-aware training remain unexplored and may further extend the Pareto frontier.

6 Conclusion

We presented PIN-Lite, a systematic model optimization framework for multimodal deepfake detectors that jointly addresses the often-overlooked dimensions of efficiency *and* explainability preservation. Through attention-aware knowledge distillation, iterative pruning, and post-training quantization, PIN-Lite reduces the baseline teacher model footprint from 57.32 MB / 15.0M parameters to 5.29 MB / 1.22M parameters ($10.83\times / 12.30\times$ size and parameter reduction) while *improving* detection accuracy from 97.37% to 98.22% on the LAV-DF benchmark.

The proposed Explainability Preservation Score (EPS) provides a principled, architecture-agnostic metric for evaluating attention-map fidelity during model optimization, revealing that standard accuracy metrics can mask critical differences in model reasoning. Our evaluation of three efficient attention variants demonstrates that Multi-Query Attention achieves the optimal accuracy–explainability trade-off (98.0% accuracy, $\text{EPS} = 0.606$, $2.71\times$ speedup), while Linear Attention—despite training-time accuracy comparable to the teacher—catastrophically fails to preserve the teacher’s attention structure ($\text{EPS} = 0.025$). This finding underscores the necessity of explainability-aware evaluation for forensic AI systems.

Comprehensive ablation studies reveal that (i) low distillation temperatures ($T \leq 2$) are optimal for binary detection tasks, (ii) the attention transfer loss is the most critical component of the distillation objective, (iii) three cross-attention layers provide the best depth–accuracy trade-off, and (iv) the distilled student is remarkably robust to pruning up to 80%.

Future work will extend PIN-Lite to larger-scale multimodal benchmarks (FakeAVCeleb, IDForge), investigate structured EPS-guided pruning, explore INT4 and mixed-precision quantization, and deploy the optimized models on physical edge hardware (NVIDIA Jetson, mobile devices) with TensorRT and ONNX Runtime acceleration.

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Declarations

- **Funding** Not applicable.
- **Conflict of interest** The authors declare no competing interests.
- **Data availability** The LAV-DF dataset used in this study is publicly available.
- **Code availability** Code will be made available upon publication.
- **Author contribution** All authors contributed to the study conception and design. All authors read and approved the final manuscript.

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